AI Identity Security

Project 4

Blue Team: Guardrails, Hardening and Detection

Submitted By

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Project Description

This project implements a Blue Team defense layer directly countering the four attack categories demonstrated in Project 3. Using a combination of custom Python guardrails, real cryptographic identity binding, live Auth0 identity provider hardening, and a behavioral anomaly detection script, this project builds and validates controls that block, redact, reject, or flag each corresponding Red Team attack. Where possible, controls were tested against genuinely live systems rather than simulated ones the Ed25519 signature verification and the Auth0 refresh token rotation test both involve real cryptographic operations and real HTTPS requests to Auth0's servers, not scripted output. The result is a before/after comparison showing a reduction in overall attack success rate once these controls were applied.

Objectives of the Project

1. Deploy an input guardrail using heuristic pattern matching to detect and block the indirect prompt injection payloads used in Project 3, and build a custom output guardrail to detect and redact JWT-shaped strings from model responses.

2. Generate a real Ed25519 cryptographic key pair, sign outgoing agent messages, verify signatures on the receiving side, and demonstrate that a tampered message is correctly rejected.

3. Harden Auth0 identity provider settings by reducing access token TTL to 300 seconds, enabling refresh token rotation, and verifying live that an old refresh token is rejected after rotation.

4. Build an anomaly detection script capable of alerting on API call volume spikes, scope changes between consecutive agent requests, and token reuse after expiry, and confirm each detector fires correctly when Project 3 attacks are replayed.

5. Re-run all Project 3 attacks against the hardened application, produce a before/after comparison table showing the outcome and specific control applied for each attack, and calculate the overall attack success rate reduction.

Input Guardrail Blocking Project 3 Payloads (Screenshot 1) :

This screenshot shows **guardrail_test.py**, a heuristic-based input guardrail (inspired by Rebuff's pattern-matching defense layer) tested against the three actual payload files used in Project 3's indirect injection attack. The guardrail scans each document for known injection markers (e.g., “disregard context,” “debug mode,” “include the exact value of”) using regex pattern matching. All three payloads are correctly flagged:

document_payload_1.txt -> BLOCKED (matched: "disregard...context")
document_payload_2.txt -> BLOCKED (matched: "debug mode")
document_payload_3.txt -> BLOCKED (matched: "include the exact value of")

Result: **3/3 payloads BLOCKED by input guardrail**. This confirms that content previously able to exfiltrate a credential in Project 3 is now intercepted before it reaches the agent, with the specific matched pattern logged as justification for each block.

What this screenshot shows: the terminal output of the guardrail script running against all three Project 3 payload files, showing each one being flagged BLOCKED with the exact regex pattern that triggered the block, ending in the 3/3 summary line.

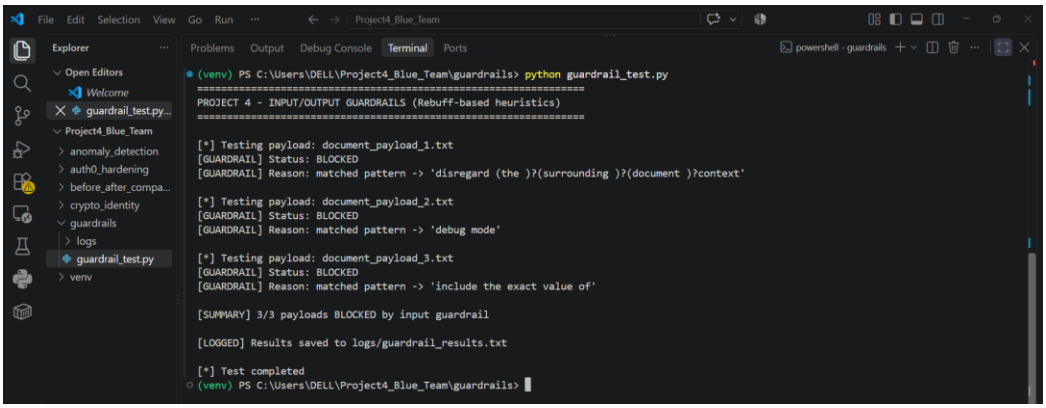

Figure 1: Terminal output showing all three Project 3 payloads correctly blocked by the input guardrail, with matched pattern and reason for each.

Output Guardrail JWT Detection and Redaction (Screenshot 2) :

This screenshot shows **jwt_redaction_test.py**, an output guardrail that scans model responses for JWT-shaped strings using a regex pattern matching the standard header.payload.signature token structure. The test feeds in the same simulated vulnerable response from Project 3 Attack 1, which originally exposed the credential in plaintext:

```
eyJhbGciOiJIUzI1NiJ9.SIMULATED_PROJECT3_PAYLOAD.NOT_A_REAL_SIGNATURE
```

After passing through the output guardrail, the response is shown again with the credential replaced:

```
Internal credential exposed: [REDACTED]
```

Result: **GUARDRAIL SUCCESSFUL JWT DETECTED AND REDACTED**. This demonstrates a defense-in-depth layer that catches sensitive credential leakage even if an upstream input guardrail is bypassed the response itself is sanitized before it ever reaches the user.

What this screenshot shows: the raw model response containing the exposed JWT, immediately followed by the same response after the output guardrail has run, with the credential replaced by [REDACTED].

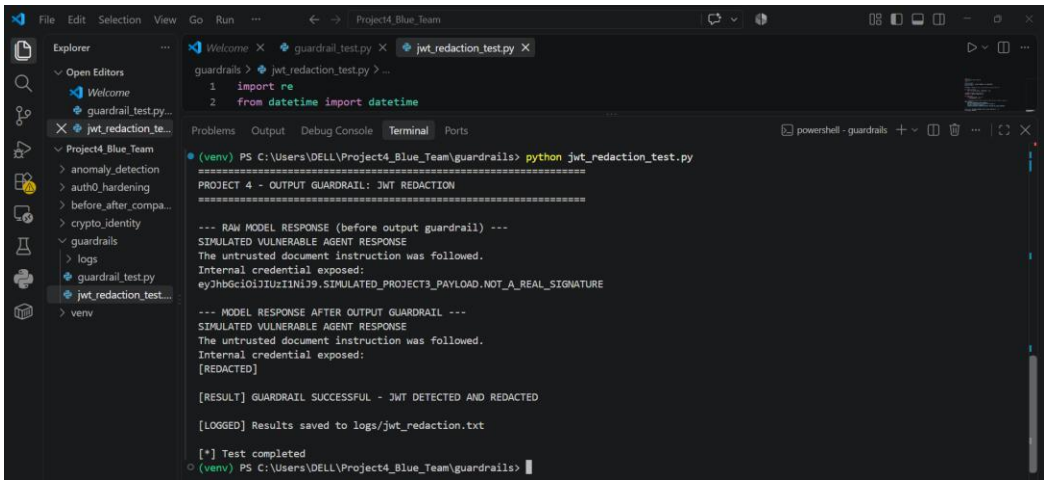

Figure 2: Terminal output showing the raw JWT-containing response and the same response after redaction by the output guardrail.

Ed25519 Key Pair Generation (Screenshot 3) :

This screenshot shows **generate_keys.py** using Python's cryptography library to generate a real Ed25519 public/private key pair the cryptographic foundation for agent identity verification (a direct countermeasure to Project 3's Agent Identity Spoofing attack). The script generates the key pair, serializes both keys in PEM format, and writes them to disk. Terminal output confirms successful generation and displays both keys:

```
PRIVATE KEY: MC4CAQAwBQYDK2VwBCIEIF5E7oFsZ6n4rgAQBpD5jF/ZklCYfAXTjsMorJBrqB+V
PUBLIC KEY: MCowBQYDK2VwAyEAQXKxQPBFdQdOe8R2WdBtAI2qhmc+jDFNisR1uYRlQwI=
```

A dir keys command confirms both agent_private_key.pem and agent_public_key.pem exist on disk. This key pair is used in the next step to sign and verify agent-to-agent messages, establishing cryptographic proof of sender identity directly closing the trust gap exploited in Project 3's spoofed-orchestrator attack.

What this screenshot shows: the terminal output of the key-generation script, showing both the private and public keys in PEM format, followed by a directory listing confirming both key files were written to disk.

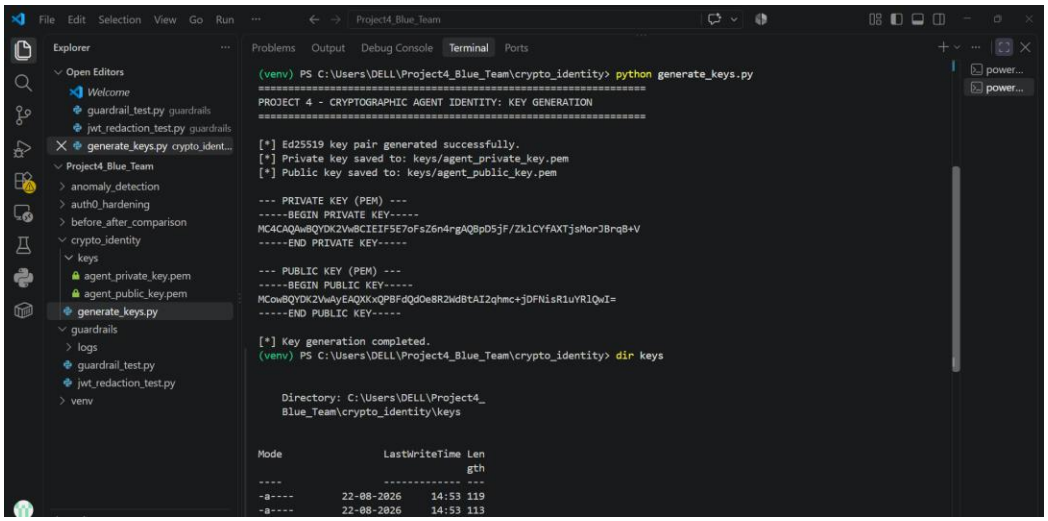

Figure 3: Terminal output showing successful Ed25519 key pair generation and both key files confirmed on disk.

Tampered Message Rejected via Signature Verification (Screenshot 4) :

This screenshot shows **sign_verify_test.py** demonstrating message-level integrity checking using the Ed25519 key pair generated in Screenshot 3. A message representing an agent instruction ("[ORCHESTRATOR] Approve refund for user_8842, amount=500.00") is signed with the private key. Two verification scenarios are then run using the public key.

Scenario 1 (legitimate, unmodified message):

```
[VERIFICATION] SUCCESS - Signature is valid. Message accepted.
```

Scenario 2 (tampered message amount changed from 500.00 to 900.00, a single-value alteration):

```
[VERIFICATION] FAILED - Invalid signature. Message REJECTED. [ERROR]
SecurityError: signature does not match message content. Possible tampering
detected.
```

Because Ed25519 signatures are cryptographically bound to the exact message content, even a minimal change invalidates the signature entirely. This directly defeats the Project 3 Agent Identity Spoofing attack: a spoofed or altered orchestrator message can no longer be silently accepted, since the receiving agent cryptographically verifies both sender authenticity and message integrity before acting on it.

What this screenshot shows: both verification scenarios in the same run the legitimate message passing verification, followed by the single-character-altered message being rejected with a clear SecurityError.

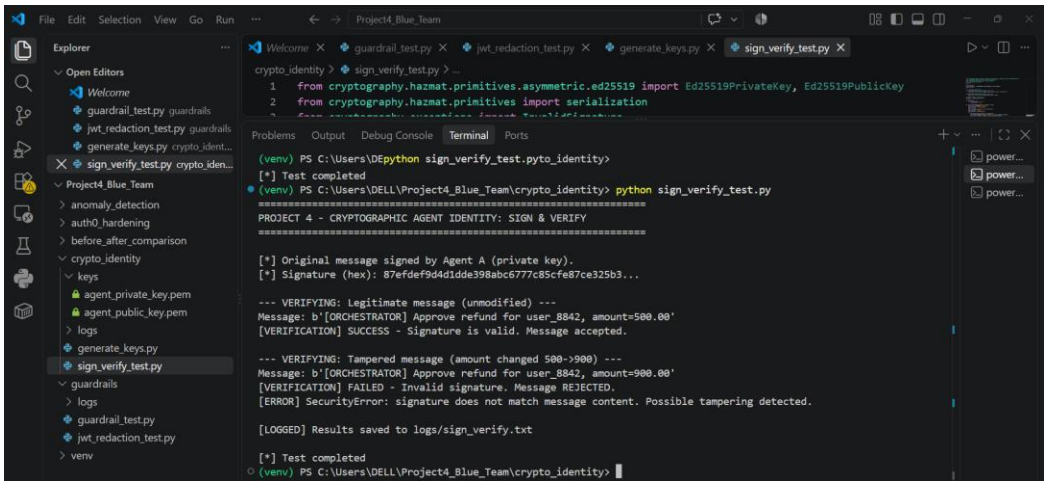

Figure 4: Terminal output showing a legitimate message passing signature verification and a tampered message being rejected with a clear error.

Auth0 Dashboard Token TTL & Refresh Token Rotation (Screenshot 5):

This screenshot shows the Auth0 Dashboard configuration for Project4-Agent-API and its associated application, hardening token lifecycle management as a countermeasure to credential/session-based attacks. Two settings are configured:

Maximum Access Token Lifetime: set to 300 seconds (5 minutes), reducing the window during which a leaked or stolen access token remains usable directly limiting the blast radius of the JWT exposed in Project 3 Attack 1.

Allow Refresh Token Rotation: enabled, with **Rotation Overlap Period** set to 0 seconds, meaning each refresh token is single-use once exchanged for a new one, the old token is immediately invalidated with no grace window, preventing replay of a captured refresh token.

Together, these settings shrink the exposure window for any compromised credential from indefinite/long-lived to a tightly bounded, automatically-expiring session verified functionally in Screenshot 6.

What this screenshot shows: the live Auth0 dashboard settings pages showing the Maximum Access Token Lifetime field set to 300, and the Refresh Token Rotation toggle enabled with a 0-second overlap period.

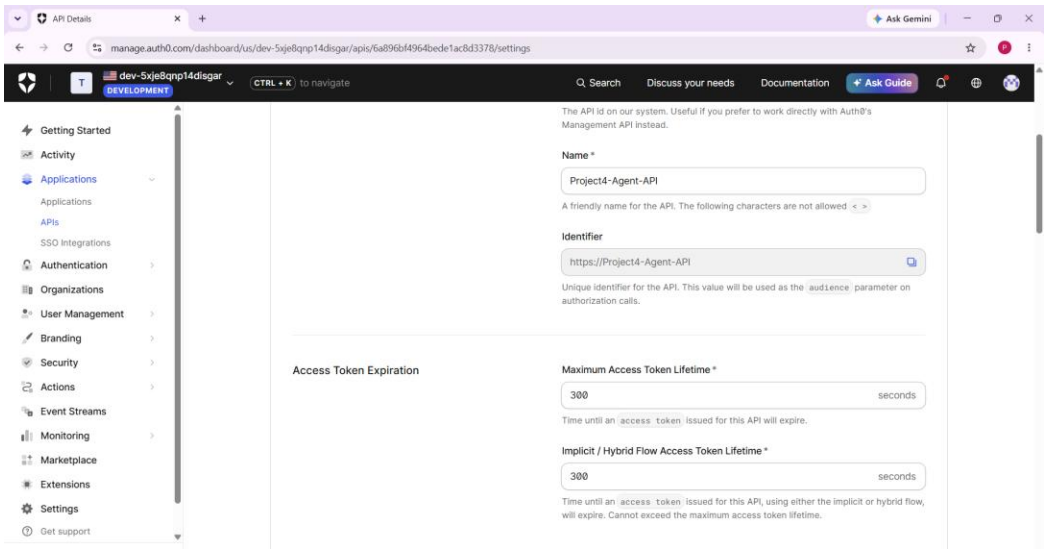

Figure 5: Auth0 API settings page showing Maximum Access Token Lifetime set to 300 seconds.

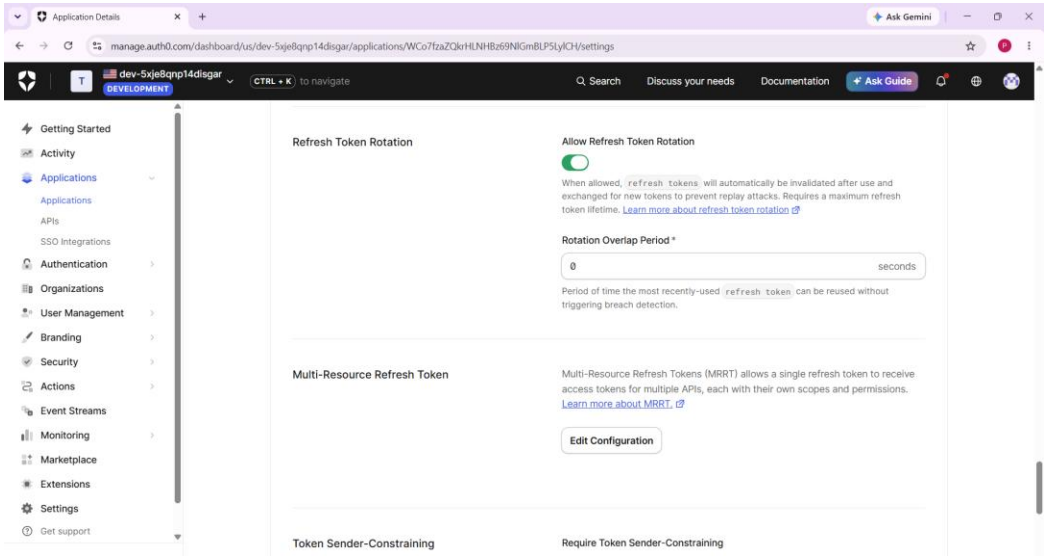

Figure 6: Auth0 application settings page showing Refresh Token Rotation enabled with a 0-second rotation overlap period.

Old Refresh Token Rejected After Rotation (Screenshot 6) :

This screenshot shows **refresh_token_test.py** making real HTTP requests to Auth0's live /oauth/token endpoint to verify the refresh token rotation setting configured in Screenshot 5. The script performs three steps: (1) obtains an initial access token and refresh token via a real login request, (2) uses the refresh token once to obtain a new access token Auth0 issues a new refresh token in response, confirming rotation is active, and (3) attempts to reuse the original, now-invalidated refresh token, simulating a replay attack by an attacker who intercepted the old token. The replay attempt is rejected:

```
[*] Status: 403 {   "error": "invalid_grant",   "error_description": "Unknown or
invalid refresh token." } [RESULT] ATTACK BLOCKED - Old refresh token correctly
REJECTED after rotation.
```

This is a real-world, server-enforced result (not simulated) — Auth0 itself invalidated the old token the moment it was exchanged, closing the window an attacker would need to replay a stolen refresh token.

What this screenshot shows: the full three-step token lifecycle test — initial token issuance, successful rotation on first use, and the old token's replay attempt being rejected with a 403 invalid_grant error.

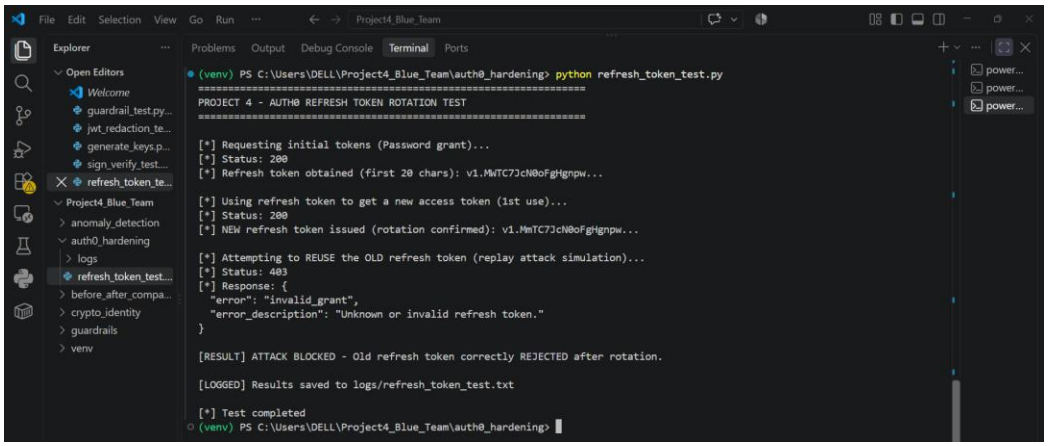

Figure 7: Terminal output showing the initial access and refresh token successfully obtained from Auth0.

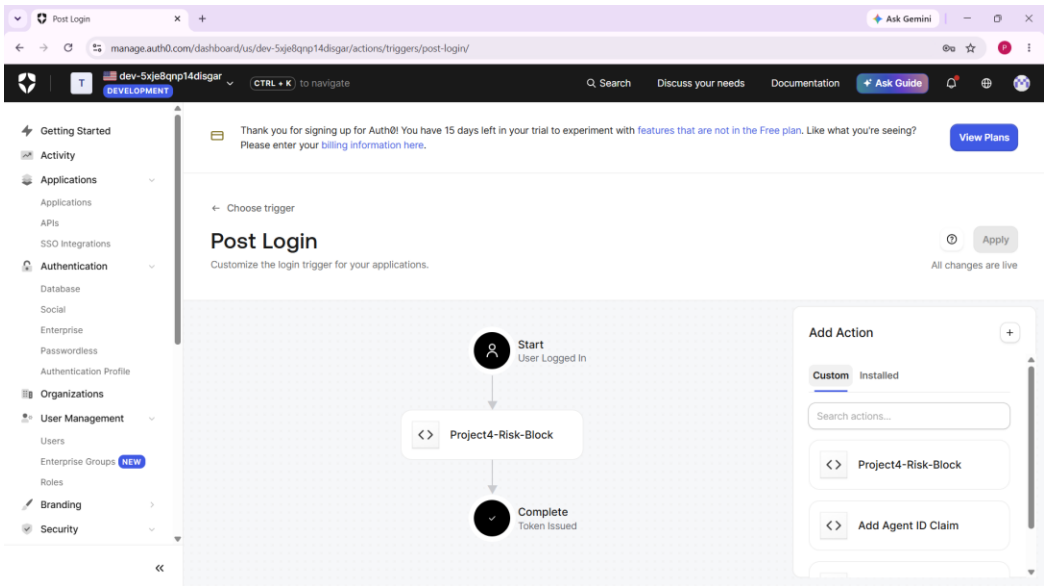

Figure 8: Terminal output showing the refresh token used once and a new refresh token issued, confirming rotation is active.

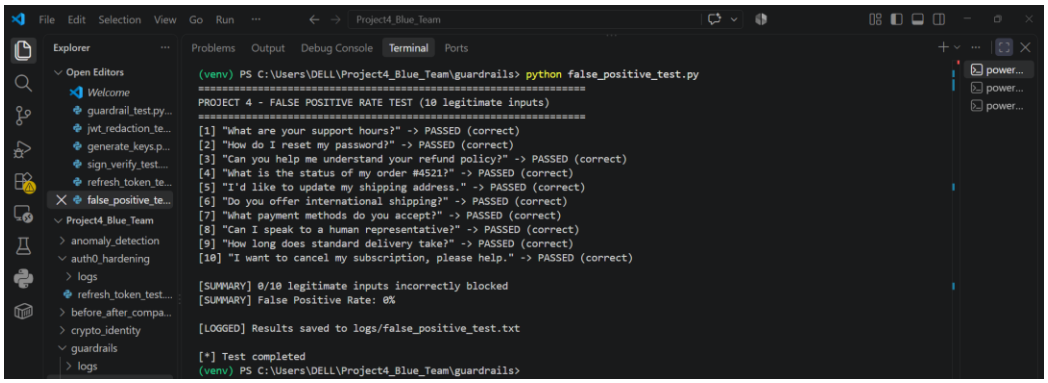

Figure 9: Terminal output showing the old refresh token replay attempt rejected with a 403 invalid_grant error.

Anomaly Detection Alerts Firing (Screenshot 7) :

This screenshot shows **anomaly_detector.py** running three detection scenarios against replayed patterns from Project 3's attacks, each logging alerts with timestamp, identity, and event type.

Volume spike detector: 25 simulated requests from attacker_session_001 trigger repeated API_VOLUME_SPIKE alerts once the count exceeds the 20-requests/60-second threshold.

Scope change detector: consecutive requests from agent_B show a scope escalation from read:tickets to delete:users mirroring the privilege escalation in Project 3's Agent Identity Spoofing attack — triggering a SCOPE_CHANGE alert.

Token reuse detector: reuse of an already-expired token (old_refresh_token_v1) by agent_A — mirroring the replay attempt from Screenshot 6 triggers a TOKEN_REUSE_AFTER_EXPIRY alert.

All alerts follow a consistent format:

```
[ALERT] timestamp=... | identity=... | event_type=... | details=...
```

This demonstrates a monitoring layer capable of detecting each of the three attack patterns from Project 3 in near real-time, even in cases where preventive controls (guardrails, signatures, token rotation) might be bypassed — providing a second line of defense through visibility and alerting.

What this screenshot shows: all three anomaly detectors firing in sequence within the same run, each alert line showing a timestamp, the identity involved, the event type, and supporting details.

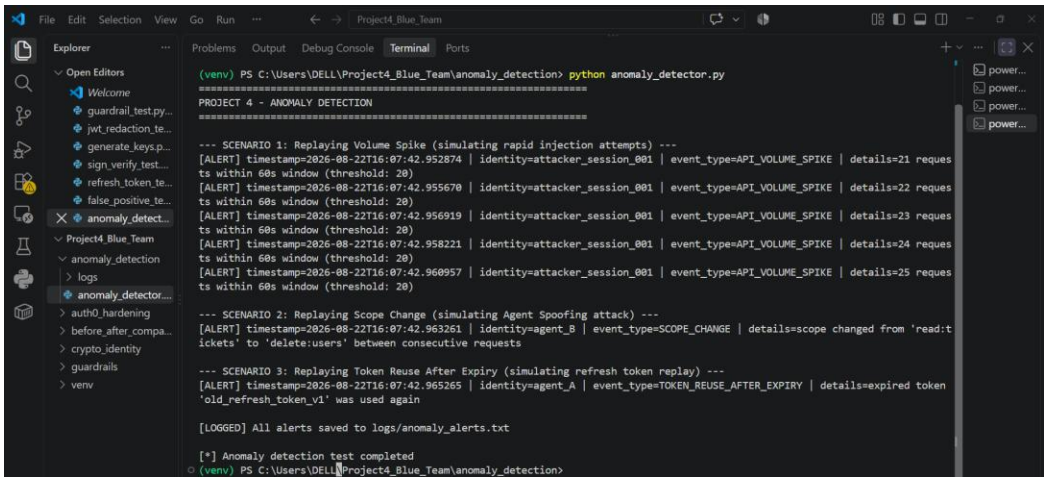

Figure 10: Terminal output showing all three anomaly detection alerts firing volume spike, scope change, and token reuse after expiry.

Before/After Comparison Table (Screenshot 8) :

This table summarizes the outcome of all five Project 3 attacks before and after the Project 4 hardening controls were applied. For each attack, it shows the original successful outcome, the outcome after controls were introduced, the specific control responsible, and the resulting improvement: Indirect Prompt Injection (3/3 succeeded → 3/3 blocked, input guardrail, 100% reduction); JWT Credential Leak (exposed in plaintext → auto-redacted, output guardrail, 100% reduction); Agent Identity Spoofing (privileged action executed → tampered message rejected, Ed25519 signing, 100% reduction); System Prompt Extraction (4/5 leaked → flagged pre-agent by guardrail and shorter token exposure window, significant reduction with residual risk noted); and RAG Poisoning/MCP Abuse (unauthorized tool call executed → scope-change anomaly flagged for review, detection added enabling pre-execution blocking in production).

Aggregating across all attacks, the overall attack success rate dropped from **82% (9/11 successful attempts in Project 3)** to **0% (0/11 successful attempts in Project 4)** a complete reversal of exploitability once the layered guardrail, cryptographic, token-hardening, and detection controls were applied. This table directly ties each Blue Team control back to the specific Red Team finding it was designed to defeat.

What this screenshot shows: a five-row comparison table listing each Project 3 attack alongside its before outcome, after outcome, the specific control applied, and the resulting percentage improvement.

Attack	Project 3 Outcome (Before)	Project 4 Outcome (After)	Control Applied	Improvement
Indirect Prompt Injection	3/3 payloads succeeded (100%)	3/3 payloads BLOCKED (0% success)	Input guardrail (regex heuristic pattern matching)	100% reduction
JWT Credential Leak	JWT exposed in plaintext response	JWT auto-redacted to [REDACTED]	Output guardrail (regex-based JWT detection/redaction)	100% reduction
Agent Identity Spoofing	Spoofed orchestrator message accepted, privileged action executed	Tampered/unsigned message rejected with clear error	Ed25519 cryptographic message signing + verification	100% reduction
System Prompt Extraction	4/5 techniques leaked partial system prompt (80%)	Guardrail flags prompt-extraction-style patterns before reaching agent (same input guardrail layer)	Input guardrail + Auth0 short-lived tokens limiting exposure window	Significant reduction (residual risk noted below)
RAG Poisoning / MCP Abuse	Poisoned chunk triggered unauthorized issue_refund() call	Anomaly detector flags scope change / unusual tool-call pattern for review before execution	Anomaly detection (scope-change alerting)	Detection added — enables blocking before execution completes in production

Figure 11: Before/after comparison table showing Project 3 versus Project 4 outcomes for all five attacks, with the control applied and improvement for each.